Agentic AI: Three Operator Cases

Voltage support, service restoration and cyber response

120 minutes with three short group exercises

IEEE ComSoc theme / Recorded research examples and operator review

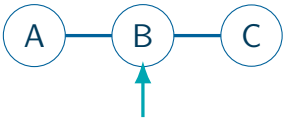
Independent educational material / IEEE ComSoc theme adaptation

Source: IEEE Communications Society. IEEE Communications Society: official resources and network presentation template. Reviewed 17 October 2026.

Three cases test three different kinds of evidence

Voltage

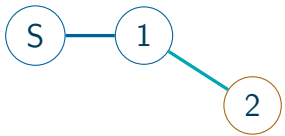
model + limits



P/Q control

Restoration

network + priority



serve critical demand

Cyber response

measurement trust



independent evidence

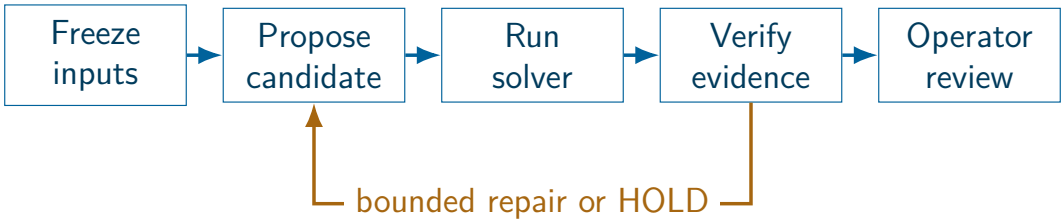
The same advisory process must answer a different engineering question in each case.

- Voltage: does the proposed control pass the required model scope?
- Restoration: who receives service? Cyber: which information can support a decision?

Source: GridGuard project teaching materials. Classroom Labs: voltage support, six-node restoration and four-bus cyber response. Existing local GridGuard workshop; recorded numerical examples and source code. 2026; inspected 18 October 2026.

An operator request becomes a checked study

The agent organizes work. Numerical tools supply engineering evidence.

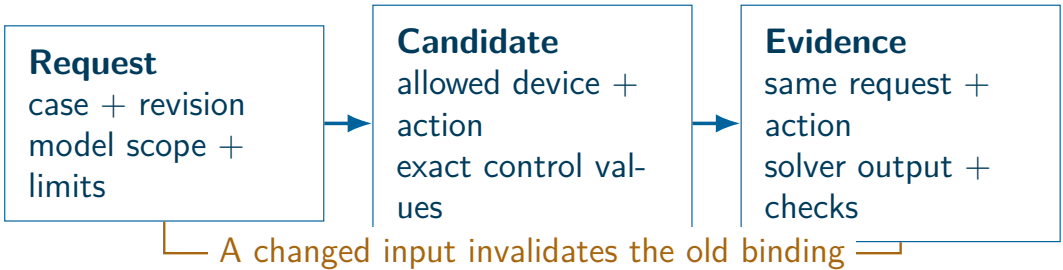


Proposed advisory workflow. Numerical checks control whether a candidate reaches human review.

- The agent interprets the request and chooses allowed tools. Solvers calculate the consequences.
- Failed or outdated evidence returns to a bounded repair step, or the study stops.

Source: GridGuard project teaching materials. Classroom Labs: voltage support, six-node restoration and four-bus cyber response. Existing local GridGuard workshop; recorded numerical examples and source code. 2026; inspected 18 October 2026.

Every result must stay attached to its inputs



Illustrative evidence identity. These version labels are teaching examples, not an operational record.

- Keep the case, model scope, data revision, candidate and limits together.
- A matching hash detects a changed packet. It does not prove that a sensor or a model is correct.

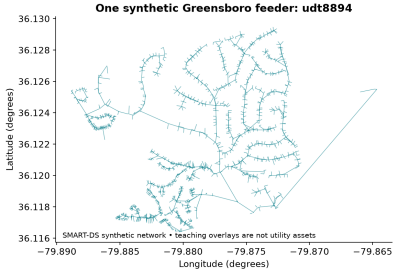
Source: GridGuard project teaching materials. Classroom Labs: voltage support, six-node restoration and four-bus cyber response. Existing local GridGuard workshop; recorded numerical examples and source code. 2026; inspected 18 October 2026.

The three exercises end with a specific next step

- Work in pairs: one person proposes the next step, the other checks the evidence.
- For each case, choose: continue the study, hold the recommendation, or request missing evidence.
- Use one number or one failed check to explain the decision. Each discussion lasts four minutes.

Source: GridGuard project teaching materials. Classroom Labs: voltage support, six-node restoration and four-bus cyber response. Existing local GridGuard workshop; recorded numerical examples and source code. 2026; inspected 18 October 2026.

Voltage support: return a command with evidence

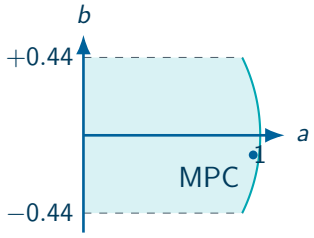


Existing SMART-DS synthetic Greensboro feeder; one teaching model, not a live utility circuit.

- Request: use available solar while staying inside voltage, current and inverter limits.
- Return an advisory command plus evidence for one synthetic feeder, uhs20_1247–udt8894.

Source: Existing GridGuard workshop / SMART-DS synthetic-feeder extraction. Synthetic Greensboro feeder uhs20_1247–udt8894: model map and declared study scope. Existing teaching snapshot reviewed 18 October 2026.

Freeze what the two control numbers mean



$p_i = aS_i, \qquad q_i = bS_i$

$0 \le a \le 1, \quad |b| \le 0.44, \quad a^2 + b^2 \le 1$

$b < 0$: reactive absorption

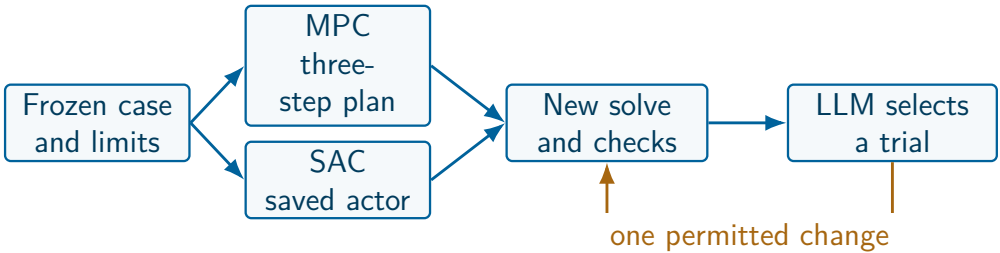
Inverter capability is one check.

Native teaching diagram of the implemented fleet capability region; the point is the recorded MPC command.

- The fleet command is (a,b): active participation and reactive injection.
- Use declared voltage/current limits; inverter capability alone does not establish feeder feasibility.

Source: GridGuard teaching materials. Continuous fleet P/Q control and classroom validation/readback helpers. Local source reviewed 18 October 2026.

Numerical tools propose; the language model selects a trial

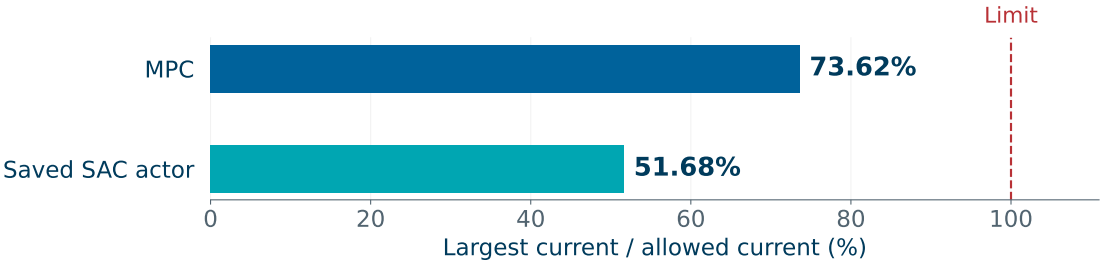


Native workflow for the existing implementation; changed commands return to the solver.

- MPC solves three forecast snapshots and proposes its first move.
- SAC proposes from saved trained weights. An LLM selects a permitted trial after numerical checks.

Source: GridGuard teaching materials. Continuous fleet P/Q control and classroom validation/readback helpers. Local source reviewed 18 October 2026.

Compare two freshly checked proposals

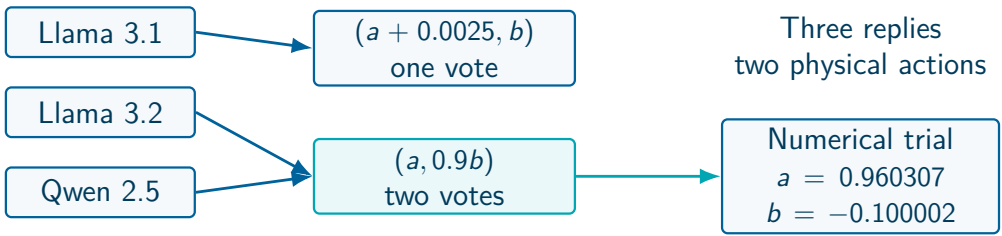


Replotted stored L2 current checks. 100% is the declared allowed-current limit.

- Fresh L2: MPC (0.960307,-0.111114); SAC (0.340066,+0.014024).
- Both pass the MV checks; current-step J is 0.031104 versus 0.436428.

Source: GridGuard teaching materials. Agentic AI for grid operators — Classroom Labs. Executed 17 October 2026.

Turn three replies into two physical trial choices

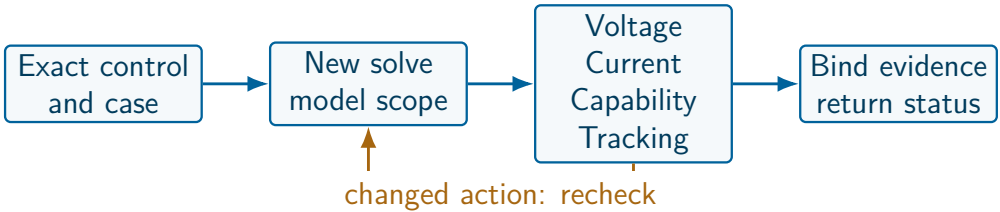


Actual archived test_01 choices; native diagram groups equal tool-and-parameter records.

- Three archived replies collapse to two distinct physical actions.
- Two votes choose reactive reduction. Selection buys one numerical trial; it does not authorize operation.

Source: GridGuard teaching materials. Archived immutable voltage contexts and actual local Llama/Qwen response records. Recorded September 2026; reviewed 18 October 2026.

Validate the changed command and bind its evidence

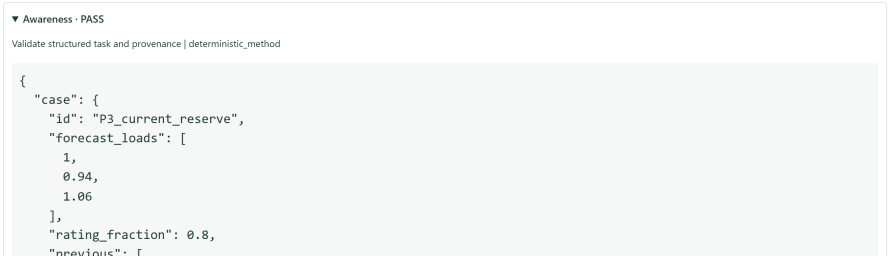


Recorded test_01 Q-reduction result, not a result for an edited classroom scenario.

- Archived Q reduction: J=0.031082, V=1.02442–1.03943 p.u., and 72.93% allowed-current use.
- These are MV-model results. Changed commands or inputs require new bound evidence.

Source: GridGuard teaching materials. Distinct final first-action numerical rechecks after bounded response selection. Recorded September 2026.

1. Inputs: the recorded OpenDSS case

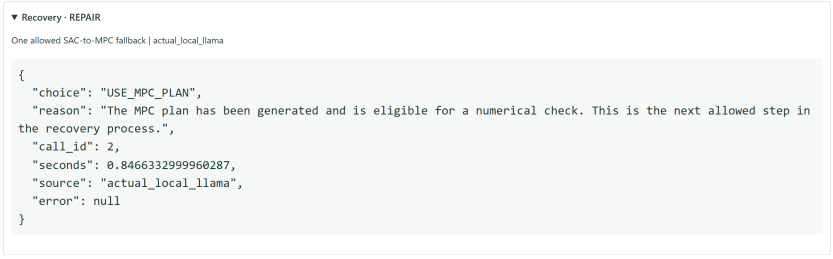


Actual dashboard Awareness excerpt, P3 OpenDSS, 27 September 2026; separate from test_01.

- P3 uses full-feeder OpenDSS, an 80% current allowance and three forecast loads.
- The model chooses allowed steps. Saved SAC and calculated MPC supply candidate controls.

Source: GridGuard teaching materials. Recorded OpenDSS voltage workflow: P3 case, actual Llama replies, dashboard and Python source. Local executed tutorial and preserved model-call journals. 27 September 2026.

2. Agent: request the MPC fallback



▼ Recovery · REPAIR

One allowed SAC-to-MPC fallback | actual_local_llama


```
{
  "choice": "USE_MPC_PLAN",
  "reason": "The MPC plan has been generated and is eligible for a numerical check. This is the next allowed step in
the recovery process.",
  "call_id": 2,
  "seconds": 0.8466332999960287,
  "source": "actual_local_llama",
  "error": null
}
```


Actual dashboard Recovery reply after the thermal rejection; no new model call was made for this slide.

- Planner chooses VERIFY_SAC_FIRST; that proposal then fails the thermal check.
- Recovery chooses USE_MPC_PLAN. The final explanation asks the operator to REVIEW_EVIDENCE.

Source: GridGuard teaching materials. Recorded OpenDSS voltage workflow: P3 case, actual Llama replies, dashboard and Python source. Local executed tutorial and preserved model-call journals. 27 September 2026.

3. Code: check the chosen command

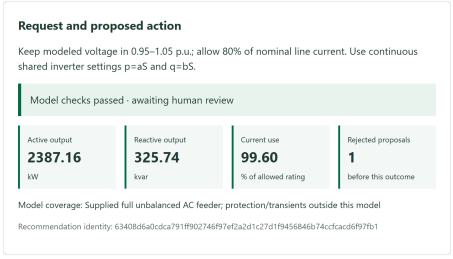
src/continuous_workflow.py, lines 231–232. Source statements; line breaks added for readability.

```
point = plant.solve(
    candidate['control'],
    load_mult=case.forecast_loads[0],
    rating_fraction=case.rating_fraction,
)
failures = verify_control(point, case)
```

- USE_MPC_PLAN selects the existing numerical candidate; the agent does not invent new setpoints.
- The loop solves that exact command with OpenDSS and calls verify_control before returning a recommendation.

Source: GridGuard teaching materials. Recorded OpenDSS voltage workflow: P3 case, actual Llama replies, dashboard and Python source. Local executed tutorial and preserved model-call journals. 27 September 2026.

4. Result: SAC fails; MPC passes



Recorded P3 first-action results. 100% denotes the declared current allowance, which is 80% of the nominal rating.

- SAC fails at 111.97% of allowed current. The MPC fallback passes at 99.60%.
- The checked proposal is P = 2387.16 kW and Q = +325.74 kvar; Vmax = 1.03503 p.u.

Source: GridGuard teaching materials. Recorded OpenDSS voltage workflow: P3 case, actual Llama replies, dashboard and Python source. Local executed tutorial and preserved model-call journals. 27 September 2026.

5. Operator: review the exact recommendation

Record acknowledgment

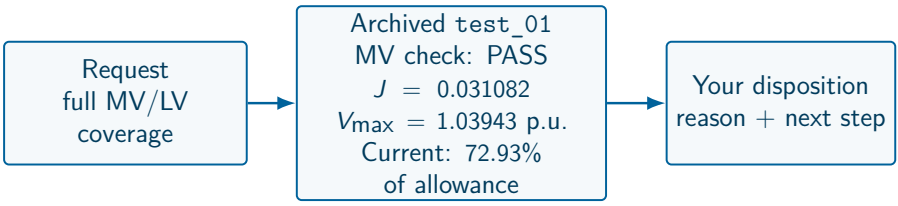
Request further review

Original operator controls, still awaiting human readback. No acknowledgment or equipment operation was performed.

- Read back P = 2387.16 kW, Q = +325.74 kvar, the 80% current allowance and full MV/LV scope.
- Then acknowledge this exact advisory record or request review. The saved human step is still PENDING.

Source: GridGuard teaching materials. Recorded OpenDSS voltage workflow: P3 case, actual Llama replies, dashboard and Python source. Local executed tutorial and preserved model-call journals. 27 September 2026.

Exercise: is this packet ready for the requested scope?



Four minutes in pairs. Metrics are recorded; the full-scope operator request is the teaching prompt.

- The request requires full MV/LV coverage. Choose a disposition, identify missing evidence, and state the next permitted step.

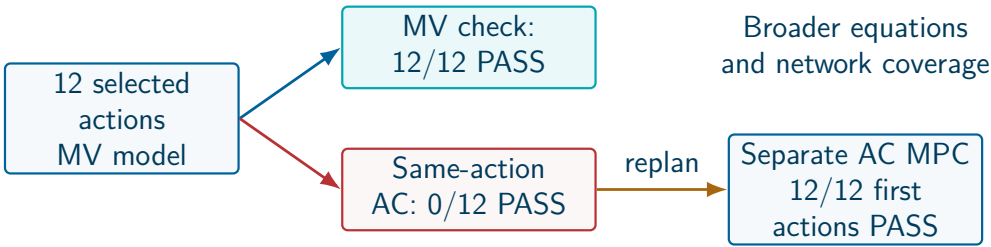
Source: GridGuard teaching materials. Distinct final first-action numerical rechecks after bounded response selection. Recorded September 2026.

Debrief: passing one model is not enough

- The request remains HOLD: MV-only evidence does not meet full MV/LV scope.
- Audit the same action in OpenDSS; if it fails, replan and verify a new command.
- Operator readback records an advisory packet; it cannot dispatch equipment.

Source: GridGuard teaching materials. Agentic AI for grid operators — Classroom Labs. Executed 17 October 2026.

Check the same actions against broader model coverage

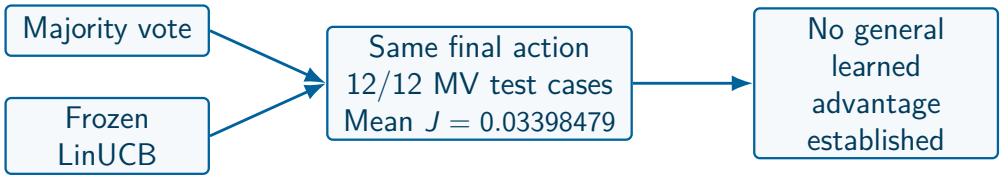


Separately recorded 12-case pilot. MV phases with ideal LV aggregation versus the supplied MV/LV model.

- On 12 held-out cases, selected MV actions pass 12/12 MV checks and 0/12 same-action AC audits.
- Broader equations and network coverage changed the result; better prose cannot repair missing model scope.

Source: GridGuard teaching materials. Frozen response-selector comparison: held-out MV test and paired full-feeder AC transfer. Protocol frozen 27 September 2026.

The historical selector ties a simpler voting rule

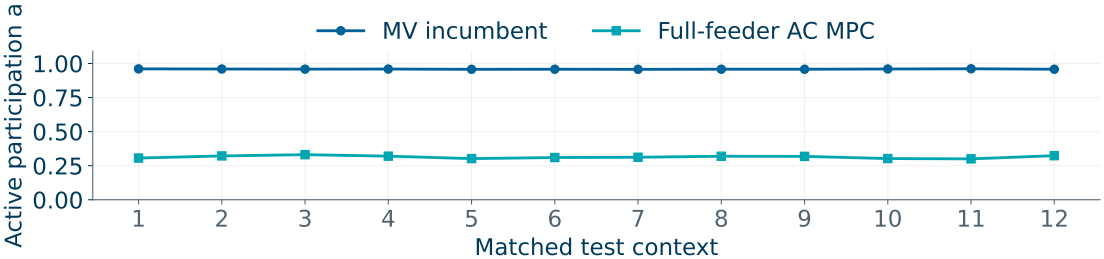


Historical single-feeder test split; lower dimensionless current-step J is preferred after the MV gate.

- Vote and frozen LinUCB tie on all 12 MV test cases: mean J=0.03398479.
- Fixed numerical refinement gives 0.03400656. This pilot does not establish learned-selection superiority.

Source: GridGuard teaching materials. Frozen response-selector comparison: held-out MV test and paired full-feeder AC transfer. Protocol frozen 27 September 2026.

Replan with the model that covers the request

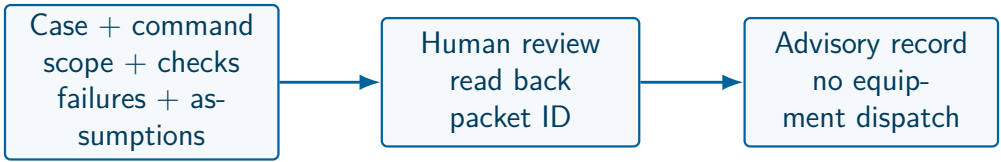


Replotted paired command records. Each curve uses a separately planned control; this is not a solver-error plot.

- Archived test_01 AC MPC proposes (0.306052,+0.027697); Vmax=1.03300 p.u. and allowed-current use=99.87%.
- All 12 AC-MPC first actions pass the AC gate; raw SAC passes 1/12.

Source: GridGuard teaching materials. Paired same-action full-feeder audit and independent OpenDSS MPC first-action plans. Recorded September 2026.

Return one scoped packet for human review



Recorded L12 review logic: acknowledgment remains a local simulation record with no equipment dispatch.

- A human acknowledges a specific checked packet.
- Scope mismatch, a wrong ID or missing operator name keeps readback blocked.

Source: GridGuard teaching materials. Agentic AI for grid operators — Classroom Labs. Executed 17 October 2026.

Restoration starts with a specific operator request

Case 2 / Service restoration

Operator request		**Study contract**		**Requested output**
Restore eligible load; keep fault 1–2 isolated.	➤	T1 + ratings priorities + scope	➤	A checked proposal for operator review

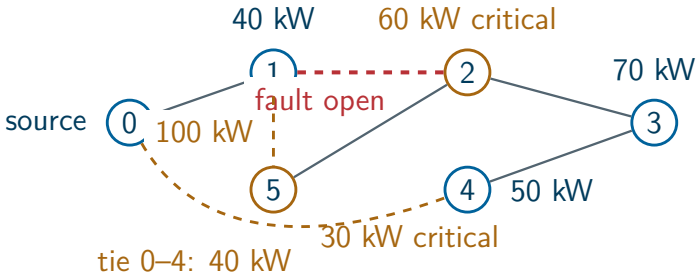
Study authority only: no automatic physical switching

Illustrative operator request for the existing six-node classroom fixture.

- Restore eligible load after isolating fault 1–2.
- Use topology T1, the stated tie ratings and critical-load priorities. Return a checked proposal for operator review.

Source: GridGuard teaching materials. Agentic AI for Grid Operators: restoration teaching chapter, frames R01–R18. Local existing workshop teaching edition. 17 October 2026 edition.

Lock the fault, topology and load priorities



Critical demand
60 + 30 = 90 kW

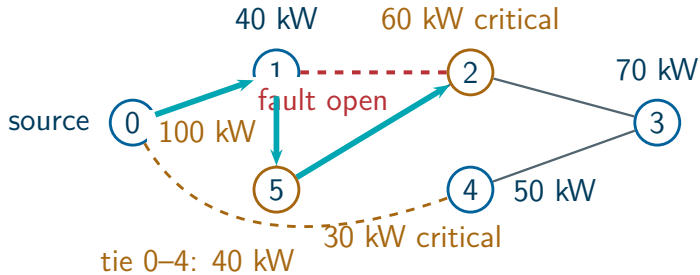
Dashed ties:
options
not commands

Original teaching schematic of the exact synthetic six-node graph; all demands are kW.

- Fault 1–2 remains unavailable throughout this study.
- Node 0 supplies the network. Nodes 2 and 5 carry 90 kW of critical demand. Dashed ties are planning options, not issued switching commands.

Source: GridGuard teaching materials. Agentic AI for Grid Operators: restoration teaching chapter, frames R01–R18. Local existing workshop teaching edition. 17 October 2026 edition.

Trace the routes before choosing a pickup plan



50 kW block does not fit a 40 kW tie

Whole load blocks no partial pickup

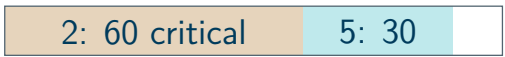
Highlighted route is a feasible path explanation, not a time-ordered switching sequence.

- The route 0–1–5–2 avoids the fault and uses the 100 kW tie.
- The other tie is only 40 kW in the first case. A 50 kW load block cannot fit through a 40 kW path alone.

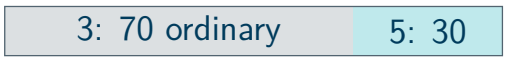
Source: GridGuard teaching materials. Classroom Labs: L6 restoration sweep and L7 changed-contract binding. Local executed Jupyter notebook. 17 October 2026 teaching edition; inspected18 October2026.

More restored kW can mean less critical service

100 kW tie budget; node 1 adds 40 kW upstream



130 total / 90 critical kW



140 total / 30 critical kW

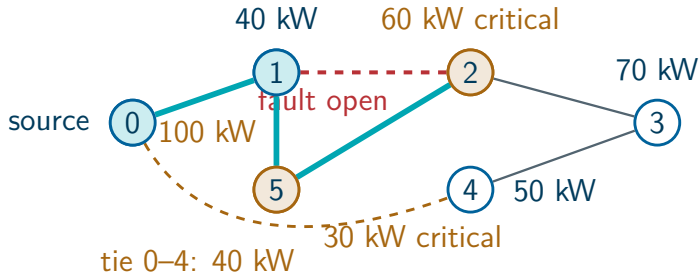
Same graph and limits; different service objectives.

Hand-checkable choices on the same graph and constraints; this is a policy comparison.

- At the 40 kW tie setting, priority weighting restores 130 kW total, including all 90 critical kW.
- Maximizing total kW restores 140 kW, but only 30 critical kW. The objective changes who receives service.

Source: GridGuard teaching materials. Agentic AI for Grid Operators: restoration teaching chapter, frames R01–R18. Local existing workshop teaching edition. 17 October 2026 edition.

Ask the solver for a plan tied to this case



Picked up:
1, 2, 5
130 kW total

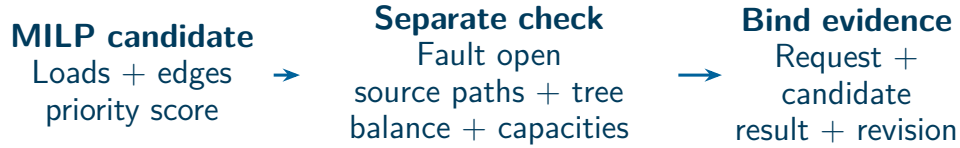
Critical service
90 kW

One declared feasible priority plan: edges 0–1, 1–5 and 5–2; fault 1–2 stays open.

- A mixed-integer linear program (MILP) chooses whole loads and connected branches.
- The 40 kW priority plan serves nodes 1, 2 and 5: 130 total, 90 critical kW. Record the chosen edges and per-branch flows.

Source: GridGuard teaching materials. Classroom Labs: L6 restoration sweep and L7 changed-contract binding. Local executed Jupyter notebook. 17 October 2026 teaching edition; inspected18 October2026.

Check capacity, source paths and the objective



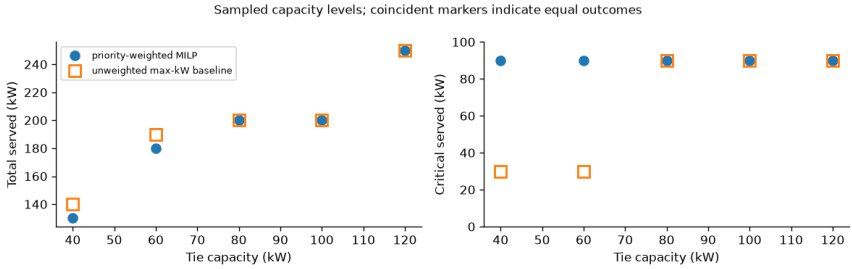
Transport check only; AC operation and switching remain unverified.

Numerical verification and evidence binding perform different jobs.

- Every picked-up load needs a source-connected path; the energized network must be a tree.
- Check branch capacities, active-power balance and fault isolation. Independent enumeration must agree with the small-case MILP objective.

Source: GridGuard teaching materials. Classroom Labs: L6 restoration sweep and L7 changed-contract binding. Local executed Jupyter notebook. 17 October 2026 teaching edition; inspected18 October2026.

A higher tie rating does not always restore another load



Existing computed Classroom Labs L6 figure; both objectives use the same feasible network.

- At both 80 and 100 kW, the weighted plan serves 200 kW total and all 90 critical kW.
- Whole load blocks and available routes create plateaus. These are sampled capacities, not a continuous response curve.

Source: GridGuard teaching materials. Classroom Labs: L6 restoration sweep and L7 changed-contract binding. Local executed Jupyter notebook. 17 October 2026 teaching edition; inspected18 October2026.

1. Inputs: the six-node restoration task

▼ Awareness | PASS: exact response


```
{
  "use_case": "restoration",
  "authorized_tool": "RESTORATION_MILP",
  "scope": "Six-node synthetic graph, active-power transport capacities, no voltage model",
  "backend": "separate_reduced_teaching_model",
  "approved_inputs": "Supplied transparent simulation fixtures; no live utility measurements"
}
```


Actual Awareness response containing the structured Planning input, 21 September 2026; tie0–4 = 100 kW.

- This recorded six-node run fixes tie 0–4 at 100 kW; the earlier example used 40 kW.
- The structured input already names RESTORATION_MILP as the allowed tool.

Source: GridGuard teaching materials. Recorded six-node restoration workflow: actual Llama routing, MILP evidence and pending human review. Local executed tutorial and preserved model-call journals. 21 September 2026.

2. Agent: choose the authorized MILP

▼ Planning | SOLVED: exact response


```
{
  "routing": {
    "choice": "RESTORATION_MILP",
    "reason": "The use case is restoration, and the authorized tool is RESTORATION_MILP, which is
suitable for load pickup and energized tree selection in a restoration scenario.",
    "call_id": 5,
    "seconds": 2.883081100007985,
    "provenance": "actual_local_llama",
    "error": null
  },
  "numerical_plan": {
```


Crop of the original Planning response: actual Llama choice, returned reason, call ID and provenance.

- Planning call 5 returns RESTORATION_MILP using the local model llama3.1:8b.
- Python checks the allowed choice and calls its fixed solver routine. Llama does not calculate the plan.

Source: GridGuard teaching materials. Recorded six-node restoration workflow: actual Llama routing, MILP evidence and pending human review. Local executed tutorial and preserved model-call journals. 21 September 2026.

3. Code: solve and check the pickup plan

src/standard_agent_workflow.py, lines 80, 83. Source statements; line breaks added for readability.

```
result = solve_restoration_milp(100.)
accepted = bool(
    result['milp']['success']
    and result['agreement']
)
```

- The actual call is solve_restoration_milp(100.): the argument sets tie 0–4 to 100 kW.
- The numerical gate requires solver success and agreement with independent enumeration.

Source: GridGuard teaching materials. Recorded six-node restoration workflow: actual Llama routing, MILP evidence and pending human review. Local executed tutorial and preserved model-call journals. 21 September 2026.

4. Result: 200 kW total, 90 kW critical

```
{
  "objective": 1010.0,
  "served_kw": 200.0,
  "critical_served_kw": 90.0,
  "pickup": [
```

Original numerical-plan excerpt. Separate verification records solver success and enumeration agreement.

- The solver serves nodes 1, 2, 3 and 5: 200 kW total, including all 90 critical kW.
- Solver success and enumeration agree. The fixed workflow returns the same numerical result without Llama.

Source: GridGuard teaching materials. Recorded six-node restoration workflow: actual Llama routing, MILP evidence and pending human review. Local executed tutorial and preserved model-call journals. 21 September 2026.

5. Operator: the record requires review

▼ Explanation | REVIEW: exact response

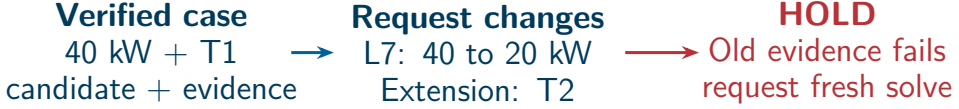

```
{
  "choice": "REVIEW",
  "reason": "The provided JSON object indicates that the numerical workflow has completed successfully,
with a served load of 200 kW and a critical served load of 90 kW. The 'passed_modeled_checks' field is
also true, indicating that the modeled checks have been passed. Therefore, the choice is to review the
results of the numerical workflow.",
  "call_id": 6,
  "seconds": 3.207767700019758,
  "provenance": "actual_local_llama",
  "error": null
}
```


Original Explanation output. The separate acknowledgment record remains PENDING, with no dispatch or equipment operation.

- Llama's Explanation call 6 returns REVIEW despite passing numerical checks. The recorded human acknowledgment remains PENDING.
- Proposed next step: inspect the scope, request missing engineering checks and review again before any separate operating authorization.

Source: GridGuard teaching materials. Recorded six-node restoration workflow: actual Llama routing, MILP evidence and pending human review. Local executed tutorial and preserved model-call journals. 21 September 2026.

A changed topology or rating makes old evidence stale



Original: **True** Changed request: **False**

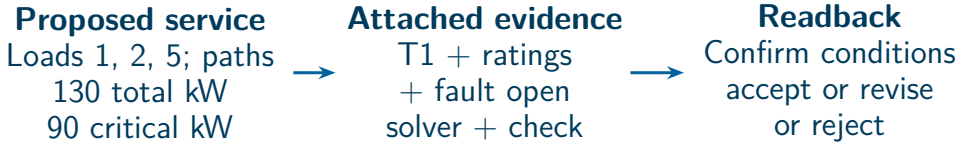
Binding failure means stale evidence;
it does not prove physical infeasibility.

L7 changes the rating/revision; the explicit topology-change test is a separately labeled teaching extension.

- A changed rating or topology revision invalidates the old evidence binding.
- L7 returns True for the original 40 kW request and False after it changes to 20 kW. HOLD and recheck; binding failure alone does not prove infeasibility.

Source: GridGuard teaching materials. Classroom Labs: L6 restoration sweep and L7 changed-contract binding. Local executed Jupyter notebook. 17 October 2026 teaching edition; inspected18 October2026.

Give the operator a packet that can be checked and read back

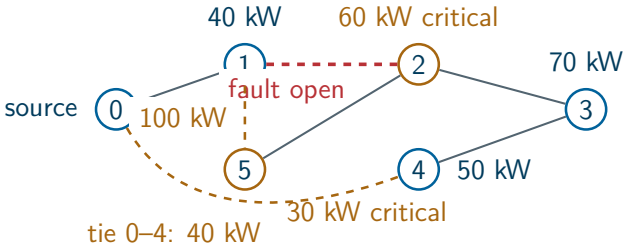


Illustrative review packet, not a switching order or a deployed control-room interface.

- Show picked-up loads, selected paths and total/critical service.
- Attach topology and rating revisions, candidate identity, solver output and independent checks. Ask the operator to confirm conditions before accepting, revising or rejecting the proposal.

Source: GridGuard teaching materials. Agentic AI for Grid Operators: restoration teaching chapter, frames R01–R18. Local existing workshop teaching edition. 17 October 2026 edition.

Exercise: choose the service policy and handle a changed case



Critical demand
$60 + 30 = 90$ kW

Dashed ties:
options
not commands

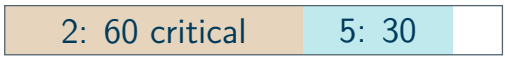
Four-minute pair exercise. Work from the map; answers appear in the debrief.

- With tie 0–4 at 40 kW, compare loads 2+5 with loads 3+5 behind the 100 kW tie.
- Add node 1's 40 kW. Report total and critical service for each choice. The topology changes after verification. Should the old packet be accepted?

Source: GridGuard teaching materials. Classroom Labs: L6 restoration sweep and L7 changed-contract binding. Local executed Jupyter notebook. 17 October 2026 teaching edition; inspected18 October2026.

Debrief: keep the priority and the evidence current

100 kW tie budget; node 1 adds 40 kW upstream



130 total / 90 critical kW



140 total / 30 critical kW

Same graph and limits; different service objectives.

Two-minute debrief: the objective and evidence validity are separate checks.

- The priority choice serves 130/90 kW total/critical; the maximum-total choice serves 140/30.
- Both can satisfy this transport model while serving different policies. After a topology change, HOLD the old packet and bind fresh evidence.

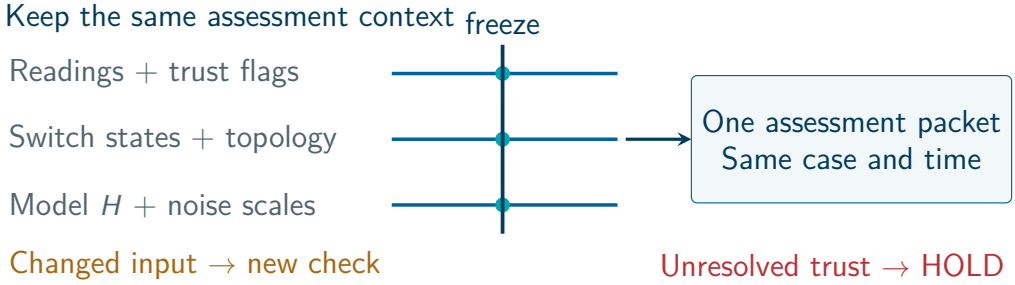
Source: GridGuard teaching materials. Classroom Labs: L6 restoration sweep and L7 changed-contract binding. Local executed Jupyter notebook. 17 October 2026 teaching edition; inspected18 October2026.

Stop at the boundary of the model and the operator's authority

- This fixture checks lossless active-power transport, source connectivity and stated branch capacities.
- It does not certify AC voltage, reactive power, losses, protection or a switching sequence.
- A reviewed proposal still needs the utility's separate authorization and operating procedure.

Source: GridGuard teaching materials. Agentic AI for Grid Operators: restoration teaching chapter, frames R01–R18. Local existing workshop teaching edition. 17 October 2026 edition.

Freeze the assessment before changing control

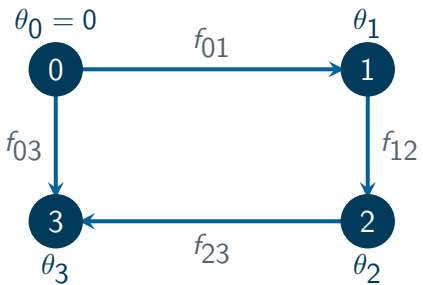


Proposed defensive assessment discipline for the four-bus teaching fixture.

- An unusual reading starts an investigation. It does not identify an attacker.
- Keep the readings, trust assumptions, topology and model in the same assessment packet.

Source: GridGuard teaching materials. Standard formulation companion: four-bus DC weighted least squares, residual tests and constrained redispatch. Local reproducible teaching fixture. 21 September 2026.

Eight readings describe three unknown angles



$x = (\theta_1, \theta_2, \theta_3)^{\mathsf{T}}$

4 flows + 3 injections + 1 angle

8 readings; $\operatorname{rank}(H) = 3$

Residual degrees: $8 - 3 = 5$

Fixed linear DC model

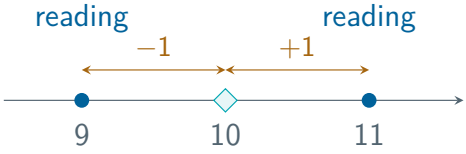
Exact topology and measurement count in the supplied four-bus fixture. Arrows define positive flow directions.

- The state contains three voltage angles. Eight readings provide five extra consistency checks.
- This is a DC model: it does not test voltage magnitudes or reactive power.

Source: GridGuard teaching materials. Standard formulation companion: four-bus DC weighted least squares, residual tests and constrained redispatch. Local reproducible teaching fixture. 21 September 2026.

WLS finds the best common explanation

Scalar illustration: equal precision



estimate $\hat{x} = 10$

Network weighted least squares

$$\hat{x} = \arg\min_x \sum_j \left(\frac{z_j - (Hx)_j}{\sigma_j} \right)^2$$

z_j: reading; $(Hx)_j$: prediction

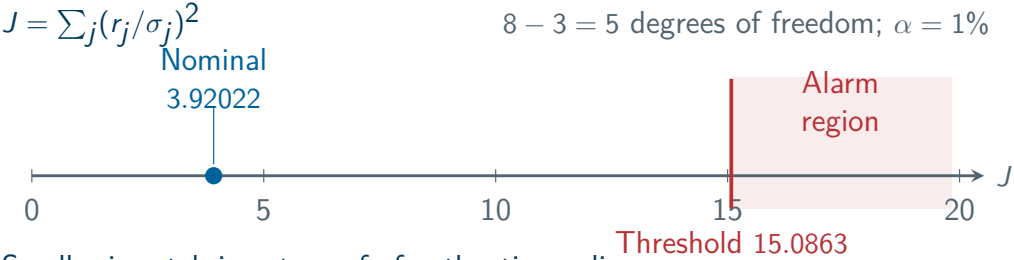
σ_j: stated noise scale

The 9–11 example explains fitting; the network calculation estimates three angles from eight readings.

- A residual is a reading minus the value predicted by the estimated state.
- WLS gives more weight to a reading with a smaller stated noise scale.

Source: GridGuard teaching materials. Standard formulation companion: four-bus DC weighted least squares, residual tests and constrained redispatch. Local reproducible teaching fixture. 21 September 2026.

A residual alarm checks consistency



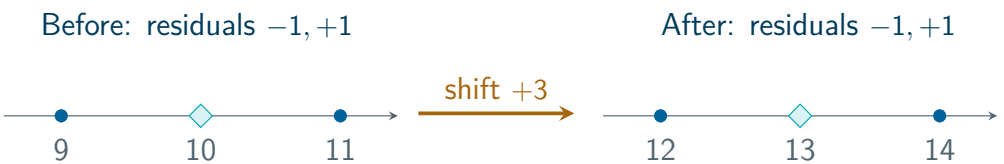
Small mismatch is not proof of authentic readings

Upper-tail consistency test with five residual degrees of freedom. A quiet alarm does not establish trustworthy data.

- Nominal J = 3.92022 is below the 15.0863 alarm threshold.
- The 1% calibration assumes the correct linear model and the stated independent Gaussian noise.

Source: GridGuard teaching materials. Standard formulation companion: four-bus DC weighted least squares, residual tests and constrained redispatch. Local reproducible teaching fixture. 21 September 2026.

Coordinated changes can leave the residual unchanged

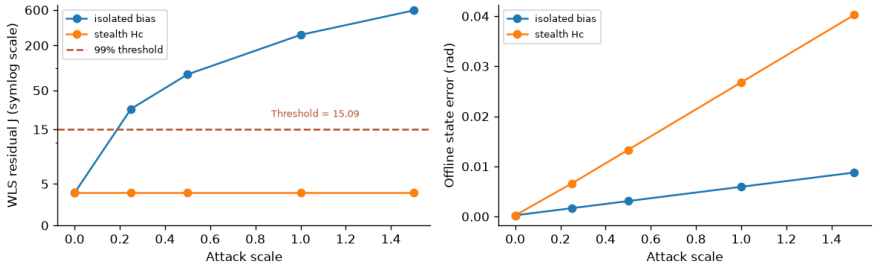


Defensive explanation of a classical state-estimation limitation. The scalar picture illustrates the cancellation.

- If readings and the inferred state move together, their mismatch can stay the same.
- The Hc result assumes a known fixed model and access to the required unprotected measurement channels.

Source: Yao Liu, Peng Ning, Michael K. Reiter. False Data Injection Attacks against State Estimation in Electric Power Grids. ACM Transactions on Information and System Security, 14(1), Article 13, 33 pages. May 2011.

A quiet residual can hide a wrong inferred state



Actual L8 sweep on the four-bus fixture. The left axis uses a symmetric-log scale; the right panel is an offline audit.

- Isolated bias raises J; the coordinated Hc change leaves J near 3.92.
- The right panel uses known synthetic truth to reveal the growing state error.

Source: GridGuard teaching materials. Classroom Labs L8–L9: cyber consistency sweep and trusted response gate. Local workshop notebook and companion lab functions. Reviewed 18 October 2026.

Independent trusted readings change the evidence

Externally supplied trust premise



3 trusted angle readings

Measurements: 3

Rank: 3

Redundancy: 0

No trusted set $\rightarrow$ HOLD

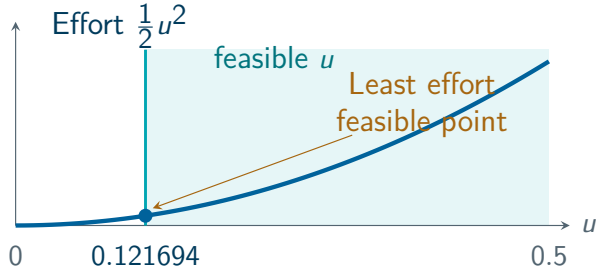
$J = 0$ supplies no authenticity proof

Exact trusted-recovery assumption: three independent angle channels, with no redundant residual test.

- The fixture supplies three separately trusted angle readings; authentication is assumed.
- Three readings for three angles leave no residual redundancy. J = 0 is not a new detection result.

Source: GridGuard teaching materials. Standard formulation companion: four-bus DC weighted least squares, residual tests and constrained redispatch. Local reproducible teaching fixture. 21 September 2026.

A small QP proposes a limited correction



All four branches:

$|\hat{f}(u)| \leq \ell - 0.01$

$0 \leq u \leq 0.5$ p.u.

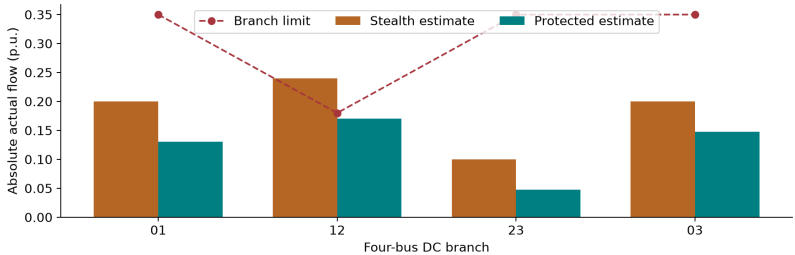
Fixed DC sensitivities

Scalar convex QP in the fixed DC model. The injection at bus 2 is balanced by reference bus 0.

- The quadratic program (QP) finds the smallest injection satisfying every predicted branch limit with a 0.01 p.u. margin.
- The trusted estimate gives u = 0.121694 p.u.; predicted flow12 becomes 0.170000 p.u.

Source: GridGuard teaching materials. Standard formulation companion: four-bus DC weighted least squares, residual tests and constrained redispatch. Local reproducible teaching fixture. 21 September 2026.

Check the consequences on every branch



Offline physical audit of the two decisions. Legend labels identify which estimate selected each action; bars show actual toy-model flows.

- Using the compromised estimate selects no correction; actual toy flow12 stays at 0.240000 p.u.
- Trusted recovery reduces it to 0.170461 p.u., below the 0.180000 p.u. rating.

Source: GridGuard teaching materials. Trusted control comparison: all-branch actual-flow audit for two four-bus DC decisions. Local workshop computed figure. Reviewed 18 October 2026.

1. Inputs: a declared four-bus DC task

▼ Awareness | PASS: exact response


```
{
  "use_case": "cyber",
  "authorized_tool": "WLS_TRUSTED_REDISPATCH",
  "scope": "Four-bus DC measurement/control example with explicitly assumed trusted sensors",
  "backend": "separate_reduced_teaching_model",
  "approved_inputs": "Supplied transparent simulation fixtures; no live utility measurements"
}
```


Actual Awareness response containing the structured Planning input, 21 September 2026.

- The structured input authorizes WLS_TRUSTED_REDISPATCH for a four-bus DC teaching case.
- Trusted sensors are assumed. The separate offline audit uses known synthetic truth.

Source: GridGuard teaching materials. Recorded four-bus cyber workflow: actual Llama routing, trusted-state correction and pending human review. Local executed tutorial and preserved model-call journals. 21 September 2026.

2. Agent: choose the authorized estimator



▼ Planning | SOLVED: exact response


```
{
  "routing": {
    "choice": "WLS_TRUSTED_REDISPATCH",
    "reason": "This tool is authorized for the given use case and scope, and it is designed for a
specific DC redispatch example with trusted measurements.",
    "call_id": 7,
    "seconds": 2.699152999994112,
    "provenance": "actual_local_llama",
    "error": null
  },
  "numerical_plan": {
```


Actual archived llama3.1:8b reply. Python executes the tool after checking the reply.

- Actual model call 7 returned WLS_TRUSTED_REDISPATCH.
- The wrapper checks that choice against the task's authorization before running the numerical tool.

Source: GridGuard teaching materials. Recorded four-bus cyber workflow: actual Llama routing, trusted-state correction and pending human review. Local executed tutorial and preserved model-call journals. 21 September 2026.

3. Code: calculate the provisional correction

src/standard_agent_workflow.py, lines 87–89. Source statements; line breaks added for readability.

```
result = run_wls_demo()
recovery = result['trusted_recovery']
control = recovery['control']
```

- The tool computes the estimate and correction; the model does not calculate the setpoint.
- The wrapper records the proposal, simulation checks and verification result separately.

Source: GridGuard teaching materials. Recorded four-bus cyber workflow: actual Llama routing, trusted-state correction and pending human review. Local executed tutorial and preserved model-call journals. 21 September 2026.

4. Result: a bounded correction with evidence

```
"flexible_injection_pu": 0.12169366726990237,
"predicted_flows_pu": [
  0.13028137485605382,
  0.16999999999999998,
  -0.046197066897314544,
```

Original injection and predicted-flow excerpt; branch order 01,12,23,03. Flow12 is the second array entry.

- The tool proposes u = 0.121694 p.u.; predicted flow12 is 0.170000 p.u.
- The offline audit gives flow12 = 0.170461 p.u. and no branch-rating violations in this fixture.

Source: GridGuard teaching materials. Recorded four-bus cyber workflow: actual Llama routing, trusted-state correction and pending human review. Local executed tutorial and preserved model-call journals. 21 September 2026.

5. Operator: resolve trust before proceeding

▼ Explanation | REVIEW: exact response


```
{
  "choice": "REVIEW",
  "reason": "The provided JSON output indicates that the numerical workflow has completed successfully,
and the results are within the expected scope. The 'passed_modeled_checks' field is set to True,
indicating that the modeled checks have been passed. Therefore, the choice is to REVIEW the results to
ensure they are accurate and within the expected range.",
  "call_id": 8,
  "seconds": 3.2306368999998085,
  "provenance": "actual_local_llama",
  "error": null
}
```


Actual archived explanation. No human approval, dispatch authorization or equipment operation is recorded.

- Actual model call 8 returned REVIEW despite the passed numerical checks.
- The record ends at PENDING acknowledgment. A person must decide what evidence or further study is needed.

Source: GridGuard teaching materials. Recorded four-bus cyber workflow: actual Llama routing, trusted-state correction and pending human review. Local executed tutorial and preserved model-call journals. 21 September 2026.

Exercise: HOLD or propose for operator review?

- A: $J = 3.92022$; measurement trust unresolved; no independent trusted replacement.
- B: $J = 277.497$; no trustworthy replacement is available.
- C: independent trust assumed; current model; all QP checks pass; $u = 0.121694$. Choose the next step and name one reason.

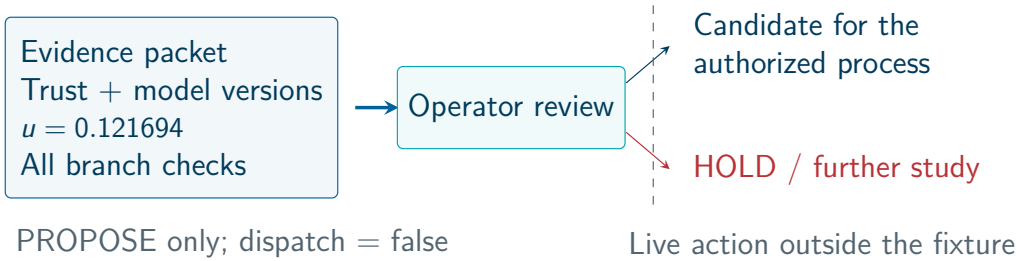
Source: GridGuard teaching materials. Classroom Labs L8–L9: cyber consistency sweep and trusted response gate. Local workshop notebook and companion lab functions. Reviewed 18 October 2026.

Debrief: consistency, trust and permission differ

- A and B: HOLD. A quiet residual and an alarm give different evidence, but neither repairs missing trust.
- C: propose for review. Passing numerical checks does not authorize an action.
- The recovery comes from trusted readings and the constrained solver; no LLM authenticates a sensor.

Source: GridGuard teaching materials. Classroom Labs L8–L9: cyber consistency sweep and trusted response gate. Local workshop notebook and companion lab functions. Reviewed 18 October 2026.

Give the operator an evidence packet



Proposed operator handoff. The classroom response is advisory and has no live control connection.

- Show the trust premise, current model, proposed action, checked limits and remaining uncertainty together.
- The fixture ends at PROPOSE with dispatch = false. Operator review is a separate decision.

Source: GridGuard teaching materials. Classroom Labs L8–L9: cyber consistency sweep and trusted response gate. Local workshop notebook and companion lab functions. Reviewed 18 October 2026.

Each failure points to a different repair

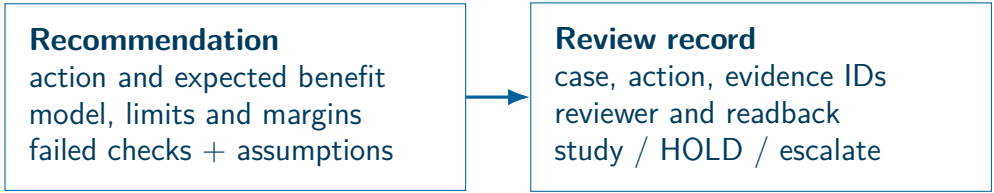
Voltage	Broader model fails	→	Replan and recheck
Restoration	Network or rating changes	→	Rerun the current case
Cyber	Trust remains unresolved	→	Independent evidence or escalate

Match the remedy to the failed evidence, rather than asking for a more confident explanation.

- Voltage needs an adequate model. Restoration needs the current network and service policy.
- Cyber response needs justified independent evidence. A new explanation alone repairs none of these.

Source: GridGuard project teaching materials. Classroom Labs: voltage support, six-node restoration and four-bus cyber response. Existing local GridGuard workshop; recorded numerical examples and source code. 2026; inspected 18 October 2026.

An operator reviews one complete recommendation packet



Study review is a separate step from authority to operate equipment.

Illustrative advisory handoff. Local acknowledgment records review and does not dispatch equipment.

- Show the proposed action, expected benefit, checked limits, failed checks and remaining uncertainty.
- Bind the reviewer's readback to the same case, candidate and evidence version.

Source: GridGuard project teaching materials. Classroom Labs: voltage support, six-node restoration and four-bus cyber response. Existing local GridGuard workshop; recorded numerical examples and source code. 2026; inspected 18 October 2026.

A useful pilot measures completed work and its errors

- Compare against the same numerical tools and a strong fixed workflow, under matched task and time budgets.
- Measure correct completed tasks, missed limits, stale evidence, reviewer effort and total elapsed time.
- Retain failures, input versions and decisions so another engineer can reproduce the finding.

Open the Full Reading Guide

Separate reading-guide PDF

Source: GridGuard project teaching materials. Classroom Labs: voltage support, six-node restoration and four-bus cyber response. Existing local GridGuard workshop; recorded numerical examples and source code. 2026; inspected 18 October 2026.